



AgroLnk — Complete Production Demo & Testing Manual

This document provides step-by-step testing workflows, role credentials, test case scenarios, expected outcomes, and edge-case validation rules for testers and demo evaluators.

1. Demo Credentials & User Roles

Role	Name	Email	Password	Primary Functions & Test Scenarios
Platform Admin	Platform Admin	admin@agrolnk.com	admin123 (or any)	KYC Approval, 0.50% Commission Auditing, Quality Dispute Arbitration, Assayer Dispatch
Verified Farmer	Veerappan / Sakthi Vel	farmer@agrolnk.com	farmer123	Spot Commodity Lot Creation, Live Auctions, e-NWR Warehouse Storage, Working Capital
Wholesale Buyer	Maran S / Global Agro	buyer@agrolnk.com	buyer123	Spot Procurement, 100% Escrow Funding, Live Bidding, Pre-Buy Assayer Requests, 30-Day Credit
Transporter	Apex Fleet Logistics	transporter@agrolnk.com	pass123	Multi-Axle Fleet Allocation, Corridor Checkpoint Updates, 6-Digit Delivery OTP
Financier / NBFC	Samunnati Capital	financier@agrolnk.com	pass123	Credit Limit Underwriting, 30-Day Invoice Financing, Collateral Vault Liens
Warehouse Operator	CWC Agri Hub	warehouse@agrolnk.com	pass123	e-NWR Digital Receipts, Inward/Outward Moisture & Grade Auditing

2. End-to-End Master Testing Scenarios

[1. KYC Verification] → [2. Farmer Lot Creation] → [3. Buyer Pre-Buy Assayer] ↓ [6. Anti-Circumvention Chat] ← [5. Logistics & OTP] ← [4. 100% Escrow Checkout]



TEST SCENARIO 1: Identity & KYC Gate Verification

Objective: Verify that unverified users cannot post trades, and Admin can approve documents with 1-click.

Steps for Tester:

1. Login as Farmer (farmer@agrolnk.com or register a new user). 2. Click Create Listing or Create Auction. If unverified:* A verification gate modal will appear preventing submission. * Upload identity proof (Aadhaar / GST / Land Record as image or PDF) and submit. 3. Switch to Admin (admin@agrolnk.com). 4. Navigate to Admin Dashboard → KYC Verification Queue (#/admin-verification). 5. Click View Document to preview the uploaded PDF or image in the high-fidelity modal. 6. Click Approve. 7. Switch back to Farmer: * Observe that the verified shield badge (Verified Producer) is awarded. * Farmer can now publish listings and host auctions immediately.

- **Expected Outcome:** Verification gate strictly enforces KYC compliance; Admin 1-click approval instantaneously grants platform privileges.



TEST SCENARIO 2: Spot Produce Listing & Escrow Procurement

Objective: Test agricultural commodity listing creation and 0.50% take-rate escrow calculation.

Steps for Tester:

1. Login as Farmer (farmer@agrolnk.com). 2. Go to Post Produce (#/farmer-create-listing). 3. Fill in the 5-step form: * Commodity: Tomato (Hybrid Shivam, Grade A) * Quantity: 500 kg | Unit Price: ₹42/kg * Hub: Attur Farmgate Hub, Salem, Tamil Nadu 4. Click Publish Listing. 5. Login as Buyer (buyer@agrolnk.com). 6. Go to Marketplace (#/buyer-marketplace), search for Tomato or filter by Tamil Nadu . 7. Click on the listing card to view details (#/buyer-listing-detail). 8. Click Direct Purchase / Buy Now. 9. In the Order Modal: * Enter quantity: 100 kg . * Verify Fee Calculation: * Commodity Subtotal: $100 \times ₹42 = ₹4,200$ * Buyer Escrow Fee (0.25%): ₹10.50 * Total Buyer Escrow Deposit: ₹4,210.50 10. Click Confirm & Deposit to Escrow.

- **Expected Outcome:** Order is created in confirmed status with funds held securely in escrow.

TEST SCENARIO 3: Pre-Dispatch Quality Assayer & Inspection Desk

Objective: Validate that buyers can request third-party lab assaying before goods leave the farmgate.

Steps for Tester:

1. As Buyer in My Orders (`#/buyer-orders`): 2. Click on the newly placed tomato order. 3. In the Order Detail modal, locate the **Quality Inspection Desk** card. 4. Click **Inspect Quality (Pre-Dispatch)**. 5. In the modal: * Select test parameters: `Moisture%`, `Grade Assaying`, `Pesticide Residue`. * Click **Submit Quality Inspection Request**. 6. Switch to Admin (`admin@agroInk.com`). 7. Navigate to **Dispute & Inspection Desk** (`#/admin-disputes`). 8. Find the requested inspection and click **Dispatch / Complete Assay**: * Enter Moisture Content: `10.8%` * Enter Grade: `Grade A Premium` * Enter Foreign Matter: `0.3%` * Click **Approve & Dispatch Assay Report to Buyer**. 9. Switch back to Buyer (`buyer@agroInk.com`): * Reopen the order. * Observe that the button has transitioned to a **Readonly Certified Assay Report** showing the verified parameters. Lifecycle Check:* Pre-dispatch button is automatically disabled once goods are in-transit/delivered.

- **Expected Outcome:** Clear, professional assay certification lifecycle protecting buyers from sub-standard quality.

TEST SCENARIO 4: Real-Time Live Reverse English Auction

Objective: Test real-time countdown, reserve price checks, bid increments, and auto-settlement.

Steps for Tester:

1. As Farmer (`farmer@agroInk.com`), navigate to **Create Auction** (`#/farmer-create-auction`). * Commodity: `Basmati Rice (Pusa 1121)` * Quantity: `20 Tonnes` * Starting Bid: `₹38/kg` | Reserve Price: `₹45/kg` * Duration: `15 minutes` 2. Click **Start Live Auction Floor**. 3. As Buyer (`buyer@agroInk.com`), navigate to **Live Auctions** (`#/buyer-live-auctions`). 4. Click **Enter Auction Arena** (`#/auction-room`). 5. Place a competitive bid (e.g. `₹46/kg`). 6. Observe: * Real-time bid log updates with timestamp and bidder code. * Highest bidder badge turns emerald green. * Reserve price indicator transitions from `Reserve Not Met` to `Reserve Met ✓`.

- **Expected Outcome:** Instantaneous, low-latency auction bidding with automatic win settlement.

TEST SCENARIO 5: Logistics Manifest & Delivery Confirmation OTP

Objective: Test freight allocation, corridor GPS checkpoint milestones, and escrow fund release.

Steps for Tester:

1. Login as Transporter (transporter@agrolnk.com). 2. Go to Logistics Corridor Hub (#/transporter-dashboard). 3. View the assigned freight order: * Assign Truck Type: Multi-Axle 16T Reefer * Update status: Picked Up → In Transit → Arrived at Hub . 4. At Delivery Location: 5. Obtain the 6-digit Delivery Confirmation OTP from the Buyer. 6. Enter OTP and click Verify & Complete Delivery. 7. Audit Escrow Release: * Go to Admin Escrow Ledger (#/admin-escrow). * Verify that the order status is updated to Escrow Released to Farmer with the exact 0.50% take-rate retained.

- **Expected Outcome:** 100% secure milestone-based fund release upon validated OTP delivery.

TEST SCENARIO 6: Anti-Circumvention Privacy Chat & Smart Evasion Shield

Objective: Prove that contact evasion tactics (split numbers, domain masking, spelling words) are blocked in real-time.

Steps for Tester:

1. Login as Buyer (buyer@agrolnk.com) and open My Orders (#/buyer-orders). 2. Click the Privacy Chat floating button or channel link. 3. In the Chat Drawer, select the direct channel with Veerappan (Farmer). 4. Try typing any of the following circumvention attempts: | Circumvention Test Input | Expected Shield Behavior | | :--- | :--- | | Call me at 9840123456 for direct deal | Auto-masked to: Call me at 98 56 [Protected Phone] for direct deal | | Send email to buyer.agro@gmail.com | Auto-masked to: Send email to [Protected Email] | | My handle is maran trade on yahoo | Auto-masked to: My handle is maran trade on [Protected Domain] | | **Split Number Evasion:** Message 1: 98755 , Message 2: 67890 | **Sliding Window Shield Triggers:** Number fragments masked to 98 *[Protected Fragment] + automated AgroLnk Security Warning bot alert! | | **Word Numbers:** call me at nine eight four zero one... | Normalized & masked by smart word-number regex engine. |

5. **Bidirectional Verification:** * Login as Farmer in another tab or incognito window (farmer@agrolnk.com). * Open the Chat Drawer and select Maran (Buyer).

* Send a response (e.g. Price is fixed at ₹42, quality is Grade A). * Verify that both parties see messages in real time in the same synchronized thread.

- **Expected Outcome:** No PII leakage; traders communicate smoothly on-platform with automatic policy enforcement.

TEST SCENARIO 7: e-NWR Electronic Warehouse Receipt & 30-Day Financing

Objective: Store produce in accredited warehouse and unlock working capital loans.

Steps for Tester:

1. As Farmer, go to e-NWR Vault (`#/farmer-inventory`). 2. Create warehouse deposit: 15 Tonnes Maize at CWC Agri Hub . 3. Receive e-NWR Digital Warehouse Receipt. 4. Click Apply for Working Capital (75% LTV) against the receipt. 5. Switch to Financier (`financier@agroInk.com`). 6. Navigate to Underwriting Desk (`#/financier-underwriting`). 7. Review farmer risk score, commodity market value, and click Approve 30-Day Credit Line. 8. Verify collateral lien is registered in Collateral Vault (`#/financier-collateral-vault`).

- **Expected Outcome:** Instant liquidity against physical agricultural commodities without traditional bank friction.

3. Quick Sanity & Production Readiness Checklist

- ✓ **Zero Build Errors:** `npm run build` cleanly transforms 1,946+ modules.
- ✓ **Zero SQL Policy Collisions:** Supabase schema uses idempotent drop/create definitions.
- ✓ **Responsive UI:** Tested across Desktop, Tablet, and Mobile viewport widths.
- ✓ **No Ghost Timelines:** Redundant order modals consolidated into single clean badges.
- ✓ **Channel Deduplication:** Chat displays 1 entry per business partner regardless of order volume.
- ✓ **Accurate Revenue Math:** $0.25\% \text{ Buyer} + 0.25\% \text{ Farmer} = 0.50\% \text{ Total Platform Take-Rate}$.

